

IPSAE notebook

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1. Environment Setup

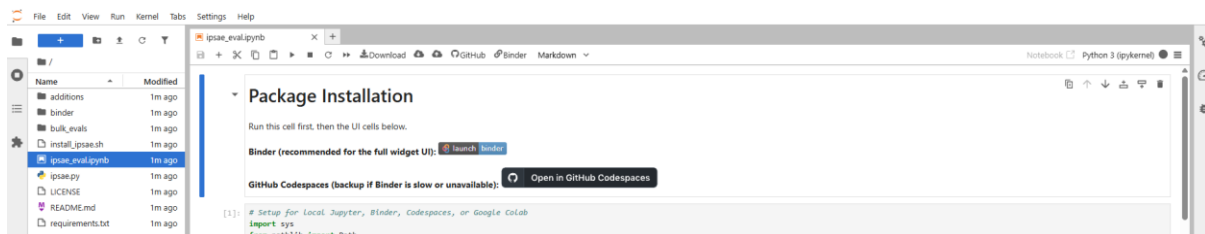
ipSAE notebook can be accessed in 2 ways from your browser:

- First option is through [Binder](#).
- Second option is through [Github Codespaces](#).

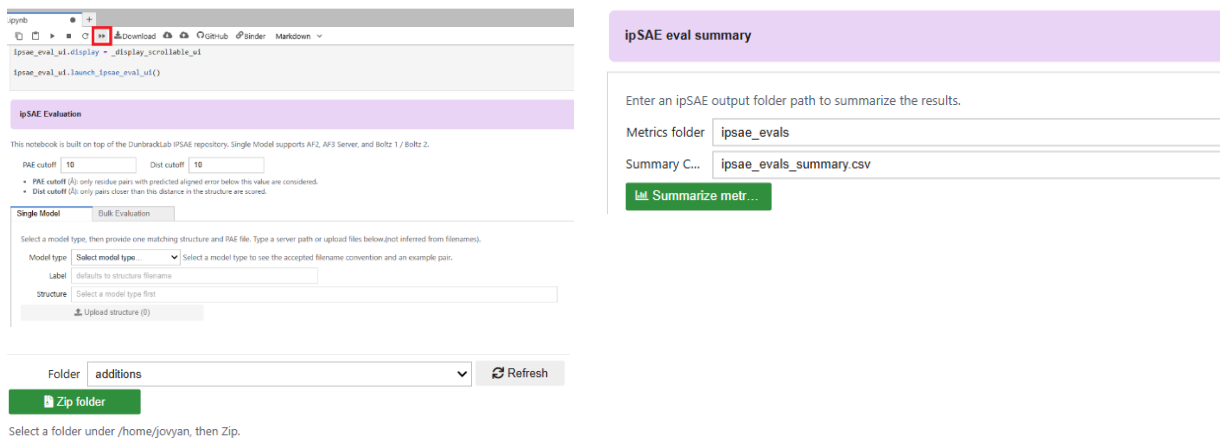
Binder Setup

Open Binder link: https://mybinder.org/v2/gh/Vincenzoos/IPSAE-notebook/main?urlpath=lab/tree/ipsae_eval.ipynb

Wait while Binder prepares the notebook and installs the required software. Once finish, you should see a Jupyter lab window and the ipsae_eval.ipynb.



Click **run all button** (highlighted in red) for all UI to render, you should see three panels namely: **ipSAE Evaluation**, **ipSAE eval summary** and **Zip folder**.



Github Codespaces Setup

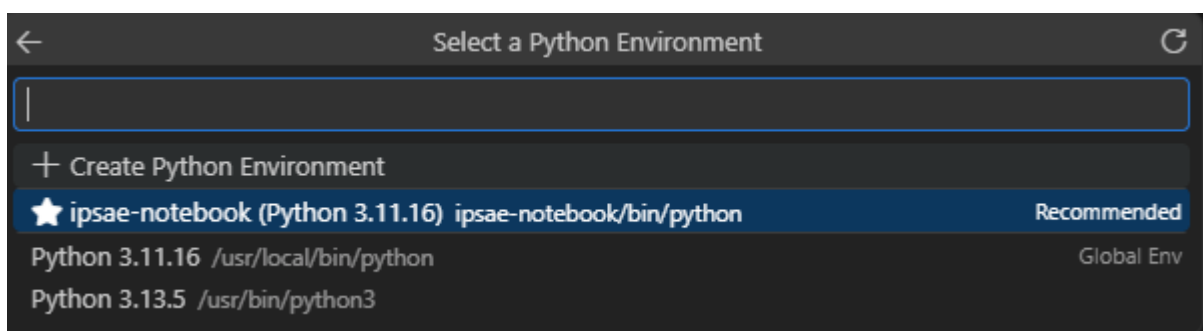
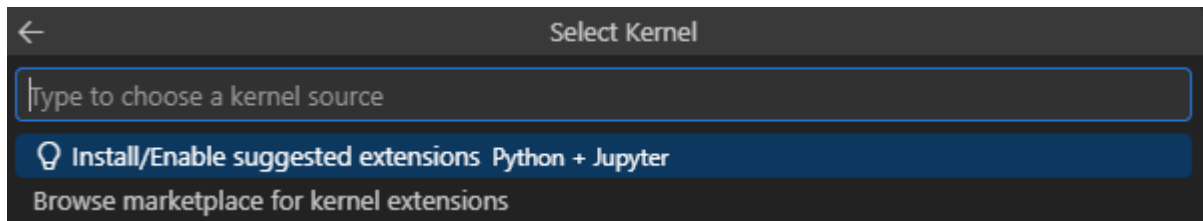
Step 1: Create Github Account

Create a GitHub account at <https://github.com/> if you do not already have one.

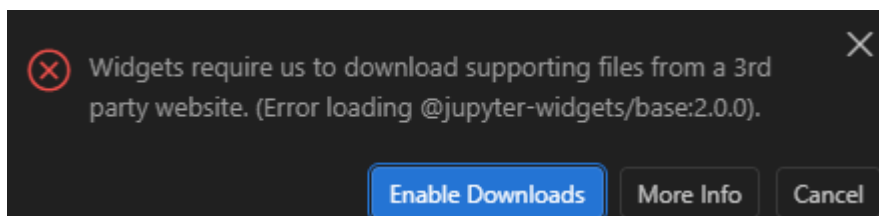
Step 2: Create Github Codespaces environment

Open codespace link: <https://codespaces.new/Vincenzoos/IPSAE-notebook?quickstart=1&editor=jupyter>, and wait for required packages to be installed.

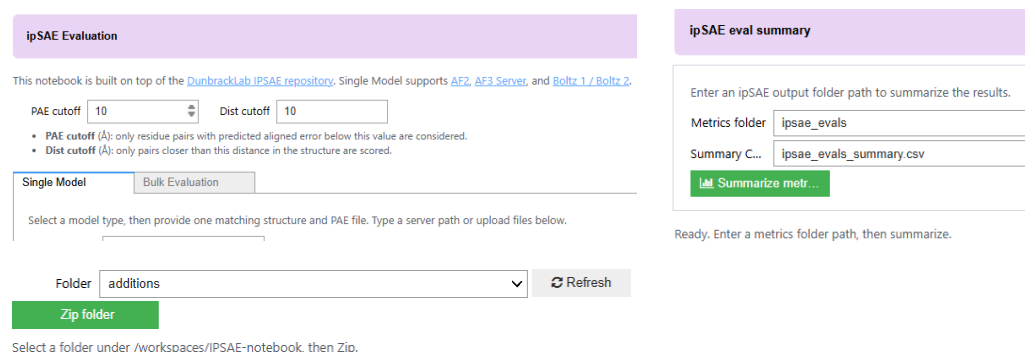
Once done, select kernel → python environment → ipsae-notebook. Install suggested extension if prompted.



When facing error loading @package-name, choose **Enable Downloads** for all.



Then click **run all** for all UI to render, you should see three panels namely: **ipSAE Evaluation**, **ipSAE eval summary** and **Zip folder**.



2. IPSAE evaluation

2.1: Configure PAE and Distance Cutoffs

PAE cutoff Dist cutoff

- **PAE cutoff** (Å): only residue pairs with predicted aligned error below this value are considered.
- **Dist cutoff** (Å): only pairs closer than this distance in the structure are scored.

Field	Description
PAE cutoff	Only residue pairs with predicted aligned error (PAE) below this limit are considered. Measured in Å (ångströms).
Dist cutoff	Only residue pairs closer than this distance in the structure are scored. Measured in Å (ångströms).

For further details about the metrics and the original ipSAE implementation, refer to:

- Dunbrack Lab ipSAE repository: <https://github.com/DunbrackLab/IPSAE/tree/main>
- ipSAE publication: <https://doi.org/10.1101/2025.02.10.637595>

Next, select an evaluation mode:

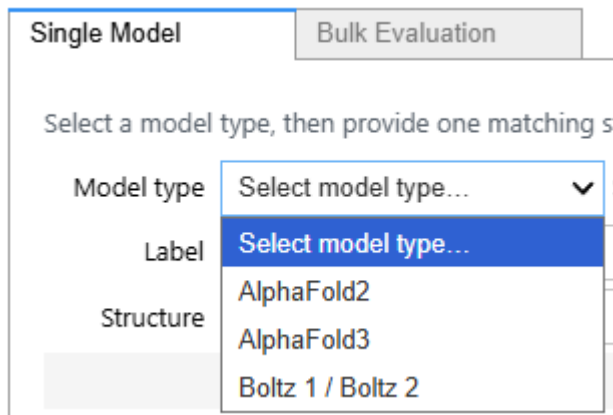
- **Single Model:** evaluate one AlphaFold2, AlphaFold3 or Boltz 1 / Boltz 2 model.
- **Bulk Evaluation:** evaluate multiple AlphaFold3 or Boltz 1 / Boltz 2 models from a folder organised as shown below.

2.2: Run the ipSAE Evaluation

Single model mode

Step 1: Select model type

Under **Model type**, select AlphaFold2, AlphaFold3 or Boltz 1 / Boltz 2.



Step 2: Upload model's file

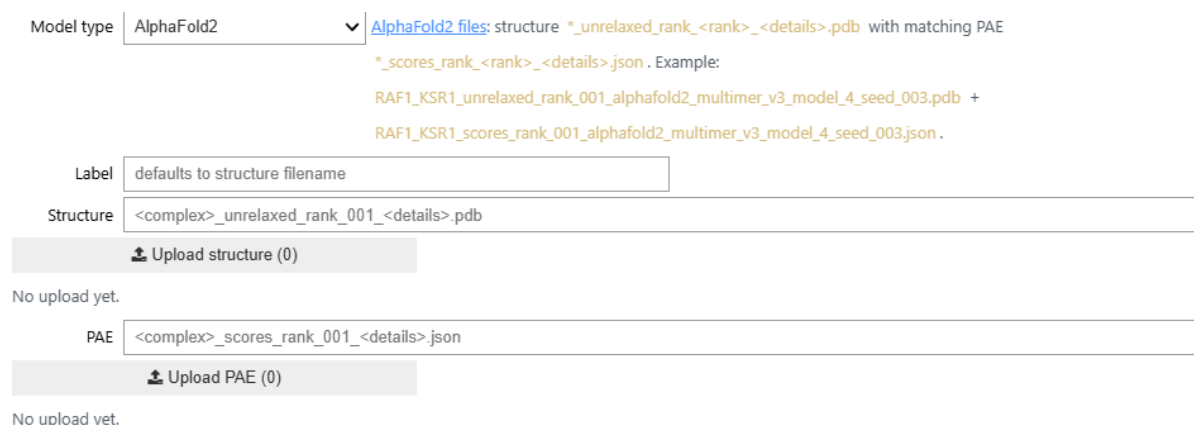
Use **Upload structure** and **Upload PAE** to select files from the same model. The filename examples below show how to identify each matching pair.

For AlphaFold2:

[AlphaFold2 files](#): upload the structure file *_unrelaxed_rank_<rank>_<details>.pdb and its matching PAE file, *_scores_rank_<rank>_<details>.json.

Example:

RAF1_KSR1_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_003.pdb +
RAF1_KSR1_scores_rank_001_alphafold2_multimer_v3_model_4_seed_003.json.



For AlphaFold3:

[AlphaFold Server files](#): upload *_model_N.cif and the matching PAE file *_full_data_N.json.

The complex name and model number (N) must match.

Example: fold_aurka_tpx2_model_0.cif + fold_aurka_tpx2_full_data_0.json.

Select a model type, then provide one matching structure and PAE file. Type a server path or upload files below.

Model type [AlphaFold Server files](#): structure *_model_N.cif with matching PAE *_full_data_N.json (same complex and N).
Example: fold_aurka_tpx2_model_0.cif + fold_aurka_tpx2_full_data_0.json .

Label

Structure

No upload yet.

PAE

No upload yet.

ipSAE Evaluation

This notebook is built on top of the [DunbrackLab ipSAE repository](#).

PAE cutoff Dist cutoff

- PAE cutoff (Å): only residue pairs with predicted aligned error below this distance in the structure
- Dist cutoff (Å): only pairs closer than this distance in the structure

Single Model

Select a model type, then provide one matching structure and PAE file.

Model type [AlphaFold Server files](#): structure *_model_N.cif with matching PAE *_full_data_N.json (same complex and N).
Example: fold_aurka_tpx2_model_0.cif + fold_aurka_tpx2_full_data_0.json .

Label

Structure

No upload yet.

PAE

No upload yet.

☒ Copy outputs to separate folder

Open

IFIT5 binder AlphaFold3... > fold_ifit5_1half_altered2_binder_I88_s481314_mpn5

Name	Date modified	Type	Size
msas	8/09/2026 3:15 PM	File folder	
templates	8/09/2026 3:15 PM	File folder	
fold_ifit5_1half_altered2_binder_I88_s481314_mpn5_model_0.cif	16/07/2026 4:33 PM	CIF File	188 KB
fold_ifit5_1half_altered2_binder_I88_s481314_mpn5_full_data_0.json	16/07/2026 4:33 PM	JSON Source File	662 KB

File name: File type:

ipSAE Evaluation

This notebook is built on top of the [DunbrackLab ipSAE repository](#).

PAE cutoff Dist cutoff

- PAE cutoff (Å): only residue pairs with predicted aligned error below this distance in the structure
- Dist cutoff (Å): only pairs closer than this distance in the structure

Single Model

Select a model type, then provide one matching structure and PAE file.

Model type [AlphaFold Server files](#): structure *_model_N.cif with matching PAE *_full_data_N.json (same complex and N).
Example: fold_aurka_tpx2_model_0.cif + fold_aurka_tpx2_full_data_0.json .

Label

Structure

Saved upload/files/fold_ifit5_1half_altered2_binder_I88_s481314_mpn5_model_0.cif

PAE

No upload yet.

☒ Copy outputs to separate folder

Output dir

Open

IFIT5 binder AlphaFo... > fold_ifit5_1half_altered2_binder_I88_s481314_mpn5

Name	Date modified	Type	Size
msas	8/09/2026 3:15 PM	File folder	
templates	8/09/2026 3:15 PM	File folder	
fold_ifit5_1half_altered2_binder_I88_s481314_mpn5_model_0.cif	16/07/2026 4:33 PM	CIF File	188 KB
fold_ifit5_1half_altered2_binder_I88_s481314_mpn5_full_data_0.json	16/07/2026 4:33 PM	JSON Source File	662 KB

File name: File type:

For Boltz 1 / Boltz 2:

[Boltz 1 / Boltz 2 files](#): upload the structure file `<complex>_model_N.pdb` or `.cif` and its PAE file, `pae_<complex>_model_N.npz`. The confidence file, `confidence_<complex>_model_N.json`, is optional. Use the same complex name and model number (N) for all files.

Example: `AURKA_TPX2_model_0.cif` + `pae_AURKA_TPX2_model_0.npz`.

Single Model

Bulk Evaluation

Select a model type, then provide one matching structure and PAE file. Type a server path or upload files below.

Model type

Boltz 1 / Boltz 2

[Boltz 1 / Boltz 2 files](#): structure `<complex>_model_N.pdb` or `.cif` with PAE `pae_<complex>_model_N.npz`.
Optional confidence `confidence_<complex>_model_N.json`. Example: `AURKA_TPX2_model_0.cif` + `pae_AURKA_TPX2_model_0.npz`.

Label

defaults to structure filename

Structure

`<complex>_model_0.pdb` or `.cif`

Upload structure (0)

No upload yet.

PAE

`pae_<complex>_model_0.npz`

Upload PAE (0)

No upload yet.

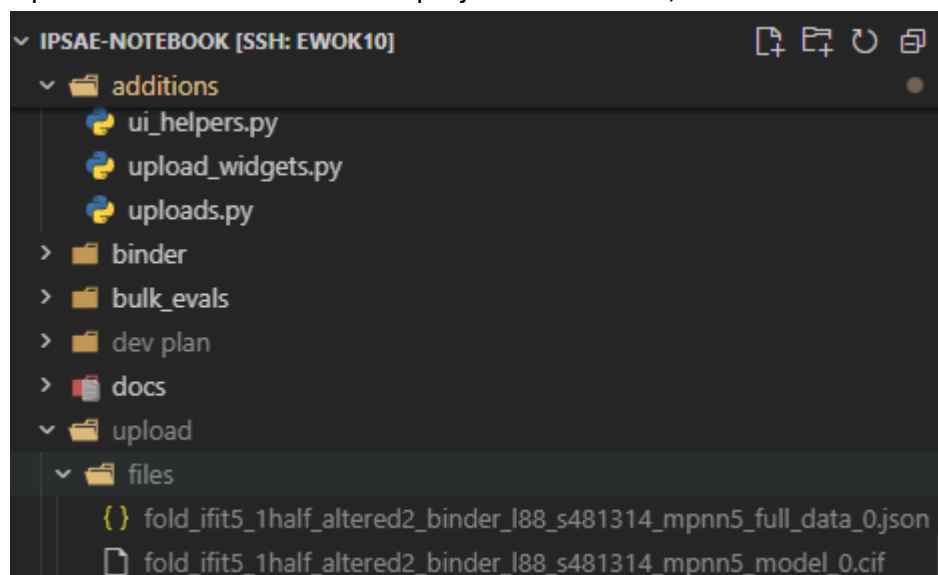
Confidence

`confidence_<complex>_model_0.json`

Upload confidence (0)

No upload yet.

Uploaded files are saved in the project's files folder, as shown below.



Step 3: Define Outputs directory

By default, ipSAE saves evaluation files alongside the structure and PAE files. Select **Copy outputs to separate folder** to keep a separate copy of the results.

☒ Copy outputs to separate folder

Output dir

In **Output dir**, enter the folder where the copied results should be saved. The default is ipsae_evals/.

Step 4: Run ipSAE

Click **Run ipSAE** to start the evaluation.

A successful run displays a confirmation message and a results table, as shown below:

SUCCESS: ipSAE completed. Outputs saved to upload/files
Copied outputs to ipsae_evals/fold_ifit5_1half_altered2_binder_l88_s481314_mpn5_model_0

SUCCESS: ipSAE completed. Outputs saved to upload/files
Copied outputs to ipsae_evals/fold_ifit5_1half_altered2_binder_l88_s481314_mpn5_model_0

	Model	Model_Type	PAE_File	Structure_File	Score_File	Eval
0	fold_ifit5_1half_altered2_binder_l88_s481314_m...	af3	upload/files/fold_ifit5_1half_altered2_binder_...	upload/files/fold_ifit5_1half_altered2_binder_...	ipsae_evals/fold_ifit5_1half_altered2_binder_...	
1	fold_ifit5_1half_altered2_binder_l88_s481314_m...	af3	upload/files/fold_ifit5_1half_altered2_binder_...	upload/files/fold_ifit5_1half_altered2_binder_...	ipsae_evals/fold_ifit5_1half_altered2_binder_...	
2	fold_ifit5_1half_altered2_binder_l88_s481314_m...	af3	upload/files/fold_ifit5_1half_altered2_binder_...	upload/files/fold_ifit5_1half_altered2_binder_...	ipsae_evals/fold_ifit5_1half_altered2_binder_...	

3 rows × 29 columns

If the evaluation fails, read the error message to identify the problem.

In this example, the structure and PAE files have different model numbers. To fix, upload a matching pair and run the evaluation again.

Single Model

Select a model type, then provide one matching structure and PAE file. Type a server path or upload files below.

Model type [AlphaFold Server files](#): structure `*_model_N.cif` with matching PAE `*_full_data_N.json` (same complex and N).
Example: `fold_aurka_tpx2_model_0.cif` + `fold_aurka_tpx2_full_data_0.json`.

Label

Structure

Saved upload/files/fold_ifit5_1half_altered2_binder_l88_s481314_mpn5_model_0.cif

PAE

Saved upload/files/fold_ifit5_1half_altered2_binder_l88_s481314_mpn5_full_data_1.json

☒ Copy outputs to separate folder

Output dir

FAILED: AF3 structure/PAE mismatch: structure and PAE refer to different model indices for the same complex ('fold_ifit5_1half_altered2_binder_l88_s481314_mpn5_model_0.cif' vs 'fold_ifit5_1half_altered2_binder_l88_s481314_mpn5_full_data_1.json').

FAILED: AF3 structure/PAE mismatch: structure and PAE refer to different model indices for the same complex ('fold_ifit5_1half_altered2_binder_l88_s481314_mpn5_model_0.cif' vs 'fold_ifit5_1half_altered2_binder_l88_s481314_mpn5_full_data_1.json').

Bulk evaluation mode

Use this mode for AlphaFold3 or Boltz 1 / Boltz 2 only, AlphaFold2 unsupported.

Step 1: Prepare the model output folder

Keep the filenames and folder organisation produced by the protein prediction tools (AF3, Boltz1/Boltz2). The examples below show the expected layouts.

Example [AlphaFold Server](#) folder layout convention

AF3_outputs/

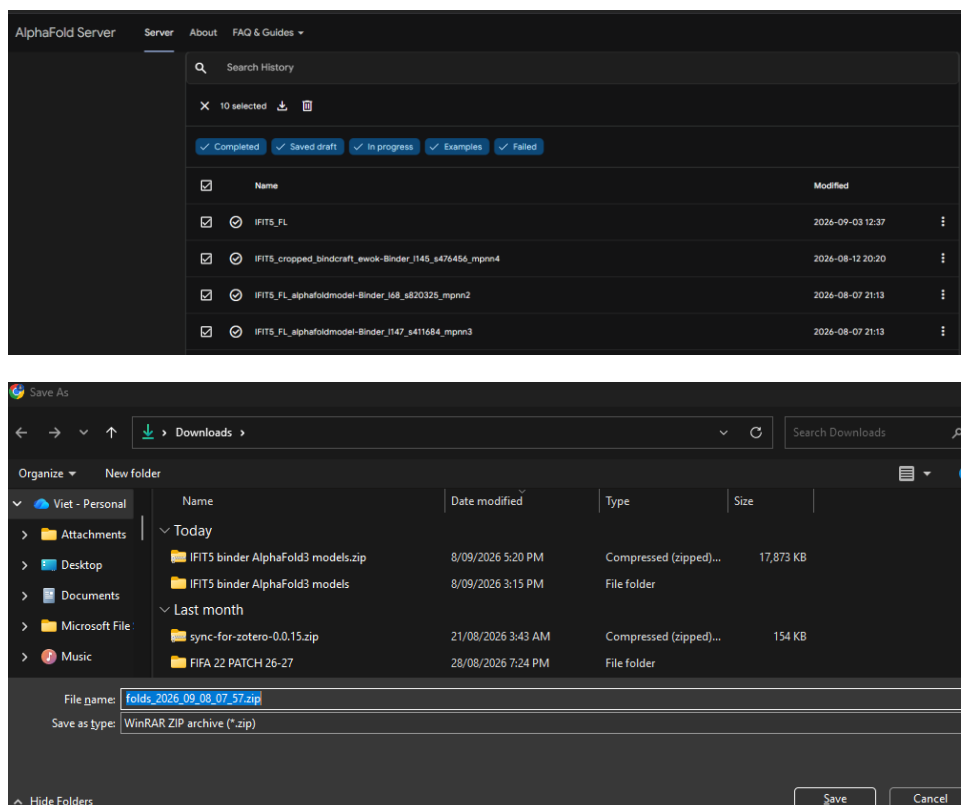
fold_binder_001/

fold_binder_001_model_0.cif

fold_binder_001_full_data_0.json

fold_binder_001_summary_confidences_0.json

On AlphaFold Server, select the models to download together. The server packages the selected results into a ZIP file (a compressed folder).



Example [Boltz 1 / Boltz 2](#) layout conventions

Boltz_outputs/

AURKA_TPX2/

AURKA_TPX2_model_0.cif

pae_AURKA_TPX2_model_0.npz

confidence_AURKA_TPX2_model_0.json

Step 2: Upload models outputs from folding prediction tools

Choose one of the following upload methods:

Method 1: Upload Widget

Click **Upload zip** in the ipSAE Eval UI panel and select your ZIP file.

Folder

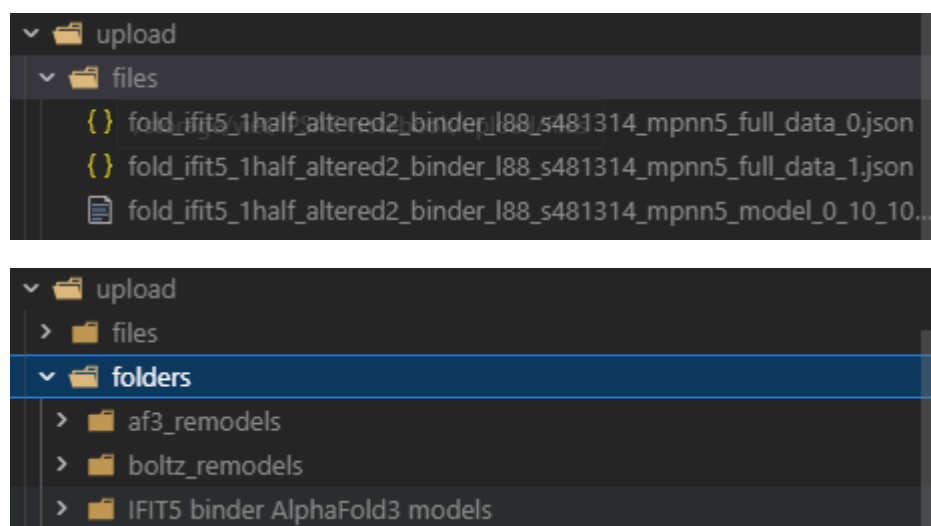
Warning: Upload limit for [AF3 Server](#) or [Boltz 1 / Boltz 2](#) zip is upto 2 GB. The Interface can hang on large transfers even though upload and evaluation jobs completed.

Zip path Extract zip

Use Jupyterlab/Github codespaces file browser instead of the upload widget for large imports.

Extracted to upload/folders/IFIT5 binder AlphaFold3 models

The uploaded ZIP file is saved in the upload/files/ folder.



The notebook automatically unpacks the ZIP file into the upload/folders/ folder.

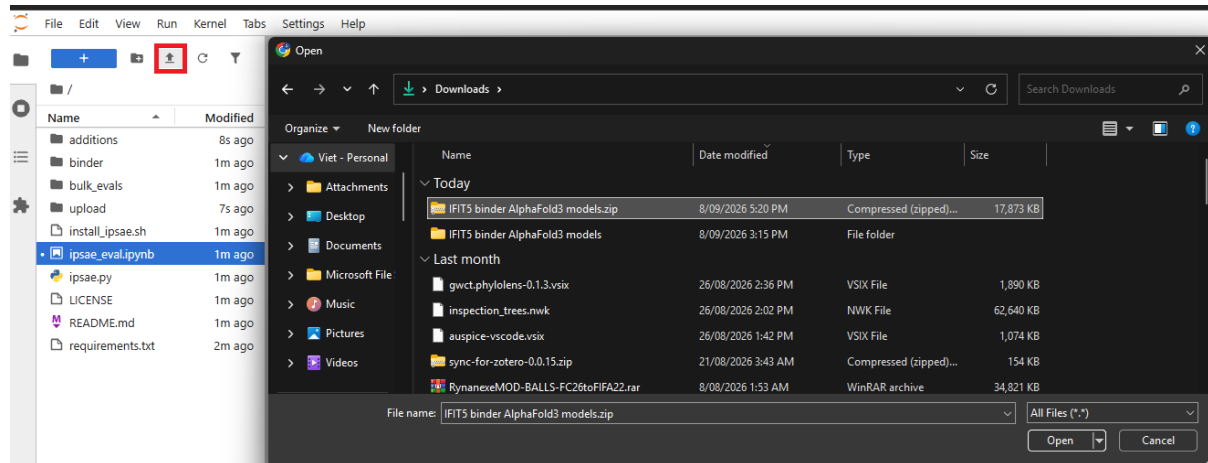
Method 2: File Browser Upload

Upload your prediction output files using the file browser in Binder or GitHub Codespaces.

On Binder

Binder uses jupyter lab environment, which doesn't allow folder upload, only file upload. Therefore, need to convert folder to zip file before upload.

In the upper-left file browser panel, click the **Upload Files** button, highlighted in red in the screenshot. Select the ZIP file containing your prediction outputs.



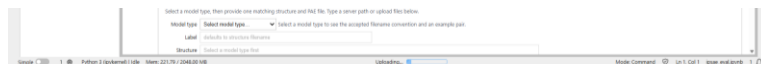
If file size warning appears, select **upload** to continue

Large file size warning

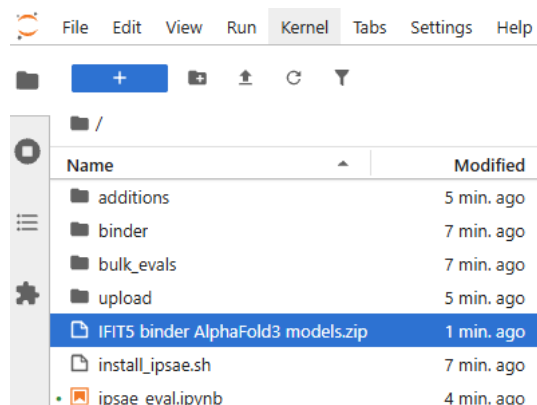
The file size is 17 MB. Do you still want to upload it?



Monitor the upload progress at the bottom centre of the window.



Once the upload is complete, the ZIP file should appear in the file browser on the left.

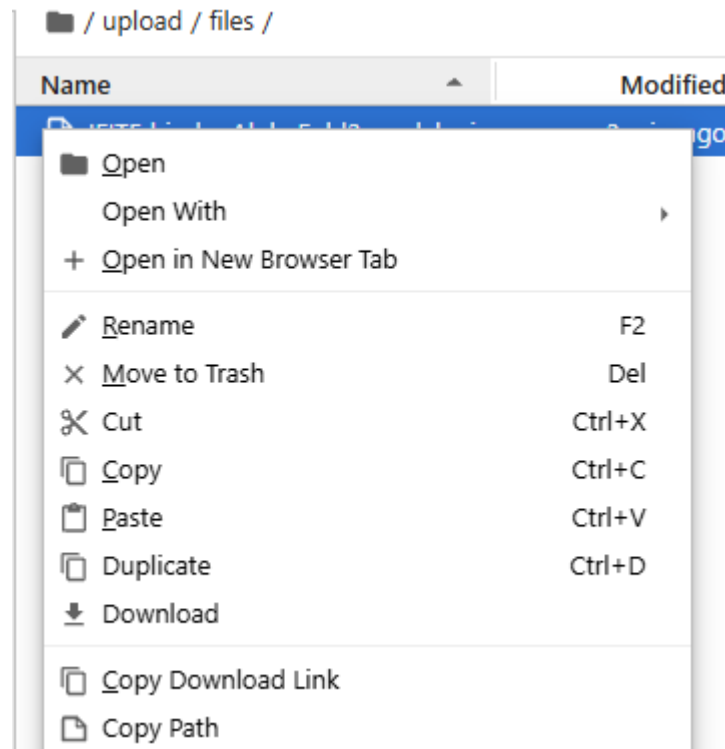


To keep your project files organised, drag the uploaded ZIP file into the *upload/files/* folder.

/ upload / files /

Name	Modified
 IFIT5 binder AlphaFold3 models.zip	3 min. ago

Right-click the ZIP file in its new location and copy its file path.



Open the **ipSAE Evaluation** panel in the ipsae_eval.ipynb notebook. Paste the copied path into the **Zip path** field, then click **Extract zip**. Wait for extraction to finish.

Warning: Upload limit for AF3 Server or Boltz 1 / Boltz 2 zip is upto 2 GB. The Interface can hang on large transfers even though upload and evaluation jobs completed.

Zip path Extract zip

Use Jupyterlab/Github codespaces file browser instead of the upload widget for large imports.

Upload zip (0)

Extracting upload/files/IFIT5 binder AlphaFold3 models.zip...

Open the extracted output folder under upload/folders/ and check that all expected model files are present.

/ ... / folders / IFIT5 binder AlphaFold3 models /

Name	Modified
fold_ifit5_1half_altered2_binder_l8...	4 min. ago
fold_ifit5_1half_altered2_binder_l8...	4 min. ago
fold_ifit5_1half_altered2_binder_l1...	4 min. ago
fold_ifit5_1half_altered2_binder_l1...	4 min. ago
fold_ifit5_1half_altered2_binder_l1...	4 min. ago
fold_ifit5_1half_altered2_binder_l1...	4 min. ago
fold_ifit5_1half_altered2_binder_l1...	4 min. ago
fold_ifit5_1half_altered2_binder_l1...	4 min. ago
fold_ifit5_1half_altered2_binder_l1...	4 min. ago
fold_ifit5_with_ifit5_1half_altered2...	4 min. ago

Next, copy the path of the extracted output folder containing all the models you want to evaluate. Paste this path into the **Folder** field of the **ipSAE Evaluation** panel.

Folder

Warning: Upload limit for AF3 Server or Boltz 1 / Boltz 2 zip is upto 2 GB. The Interface can hang on large transfers even though upload and evaluation jobs completed.

Zip path

Use Jupyterlab/Github codespaces file browser instead of the upload widget for large imports.

Extracting upload/files/IFIT5 binder AlphaFold3 models.zip...

Model index

Which ranked prediction to evaluate (files named *_model1_N / *_fu11_data_N). Defaults to 0 for best model for both AF3 Server and Boltz 1 / Boltz 2.

Detected AlphaFold3. Found 10/10 ready model(s) for index 0. Review the table, then run ipSAE.

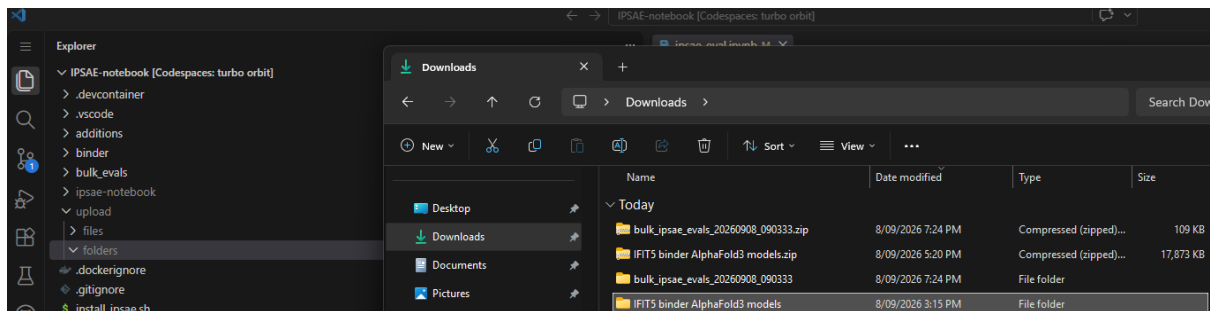
Detected AlphaFold3. Found 10/10 ready model(s) for index 0. Review the table, then run ipSAE.

Provided the original AF3 or Boltz folder structure have been satisfied, the panel should be able to detect the models in that folder. Check that the number of detected models matches the number you expect to evaluate.

Note: On Binder, the interface may appear unresponsive during file upload, ZIP extraction, or ipSAE evaluation. This does not necessarily mean that the operation has failed. Check the relevant upload, extraction, or results folder for the expected files rather than relying only on the progress indicator. Confirm that all expected files are present before proceeding.

On Codespaces

Unzip the AF3 or Boltz output folder on your computer. In Codespaces, drag the extracted folder into upload/folders/ in the **Explorer** panel.



Monitor the upload progress in the status bar at the bottom of the VS Code window. Wait until the upload is complete before searching for models.



Then, load the folder to into the **ipSAE Evaluation** panel by right click → copy relative path, and paste copied path into **Folder** field.

Once the upload is complete, select the **Model index** you want to evaluate. The default value is **0** (for best model). Next, click **Find models** to search the selected folder. Review the list of detected models and confirm that all the models you intend to evaluate are included.

Folder

Warning: Upload limit for **AF3 Server** or **Boltz 1 / Boltz 2** zip is upto 2 GB. The Interface can hang on large transfers even though upload and evaluation jobs completed.

Zip path

Extract zip

Use Jupyterlab/Github codespaces file browser instead of the upload widget for large imports.

Upload zip (0)

If using file browser instead of the upload widget for uploading **AF3 Server** or **Boltz 1 / Boltz 2** zip file(upto 2 GB), then paste its path above and click Extract zip.

Model index

Which ranked prediction to evaluate (files named ***.model.N** ***.full_data.N**). Defaults to 0 for best model for both **AF3 Server** and **Boltz 1 / Boltz 2**.

Find models

Run ipSAE

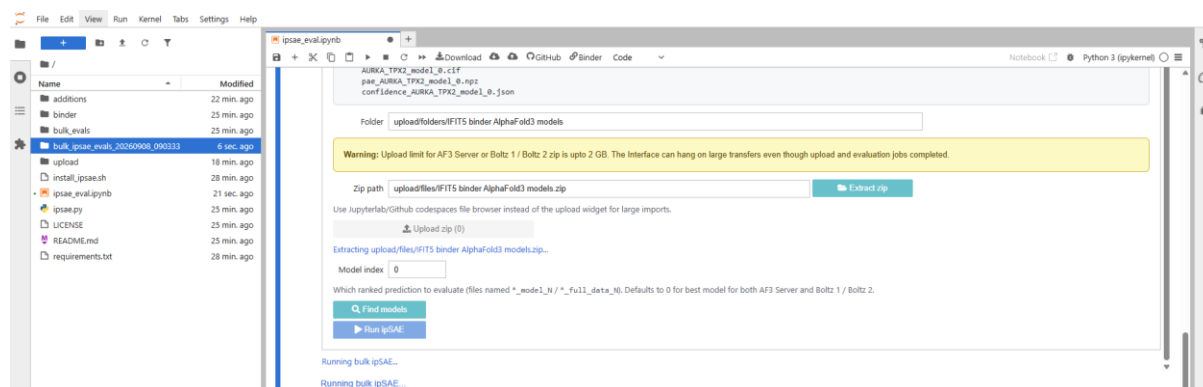
Detected AlphaFold3. Found 10/10 ready model(s) for index 0. Review the table, then run ipSAE.

Type	Model	Folder	ModelIndex	Structure
0	af3 fold_ifit5_1half_altered2_binder_I130_s734871_...	upload/folders/IFIT5 binder AlphaFold3 models/...	0	fold_ifit5_1half_altered2_binder_I130_s734871_... fold_ifit5_1half_altered2_b...
1	af3 fold_ifit5_1half_altered2_binder_I130_s734871_...	upload/folders/IFIT5 binder AlphaFold3 models/...	0	fold_ifit5_1half_altered2_binder_I130_s734871_... fold_ifit5_1half_altered2_b...
2	af3 fold_ifit5_1half_altered2_binder_I136_s450785_...	upload/folders/IFIT5 binder AlphaFold3 models/...	0	fold_ifit5_1half_altered2_binder_I136_s450785_... fold_ifit5_1half_altered2_b...
3	af3 fold_ifit5_1half_altered2_binder_I136_s450785_...	upload/folders/IFIT5 binder AlphaFold3 models/...	0	fold_ifit5_1half_altered2_binder_I136_s450785_... fold_ifit5_1half_altered2_b...

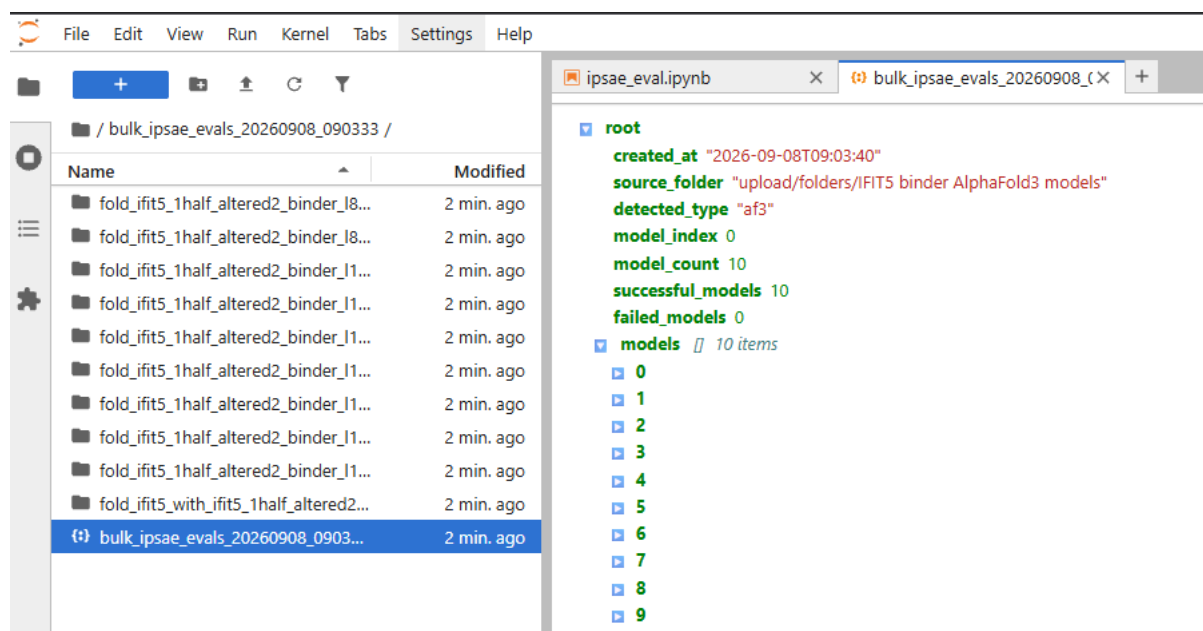
Step 3: Run ipSAE

Once the models have been detected and the required settings have been configured, run the **ipSAE evaluation**.

After the evaluation is complete, the output files will be generated in output folder with prefix: `bulk_ipsae_evals_*`



Verify if the run successful by checking the JSON log with prefix: `bul_ipsae_evals_*.json`, if none failed_models found, the evaluation was successful.



For each evaluated model, ipSAE generates three main output files:

- `*_10_10.txt` — summary of the calculated scores.
- `*_10_10_byres.txt` — per-residue evaluation results.
- `*_10_10.pml` — PyMOL script for visualising the evaluated structure and interface.

3. IPSAE Results Summary

The individual score files of an ipSAE output can be combined into a single .csv summary file. This provides a more convenient overview for comparing the evaluated models.

In the **ipSAE Eval Summary** panel, copy the path of the folder containing the generated ipSAE results and paste it into the **Metrics folder** field. Once summarized, the panel will confirm that the summary file has been created and indicate how many score files and models were included in the summary.

ipSAE eval summary

Enter an ipSAE output folder path to summarize the results.

Metrics folder:

Summary C...:

Summarize metr...

SUCCESS: summarized 310 score file(s) from 11 model(s). Saved to bulk_ipsae_evals_20260908_090333/ipsae_evals_summary.csv

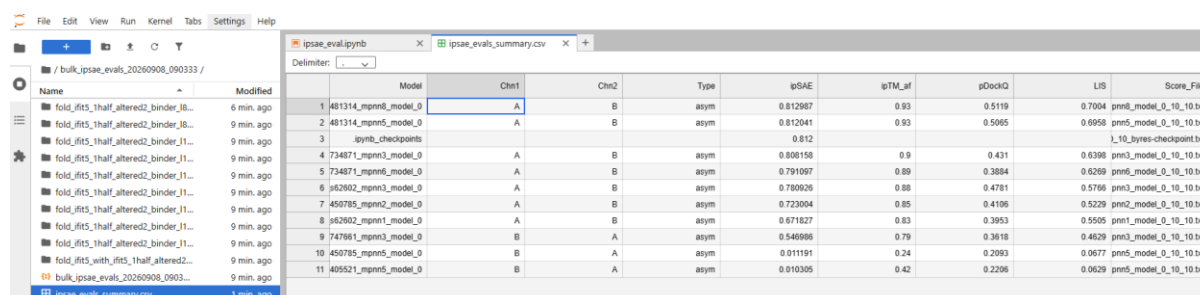
SUCCESS: summarized 310 score file(s) from 11 model(s). Saved to bulk_ipsae_evals_20260908_090333/ipsae_evals_summary.csv

Return to the ipSAE output folder and open the generated .csv summary file. This file provides an overview of important metrics, including ipSAE, ipTM, pDockQ, and LIS.

Field	Description
ipsae	ipSAE value for given PAE cutoff and d0 determined by number of residues in 2nd chain with PAE<cutoff
ipTM_af	AlphaFold ipTM values. For AF2, this is for whole complex from json file. For AF3, this is symmetric pairwise value from summary json file.
pDockQ	score from pLDDTs from Bryant, Pozotti, and Elofsson. https://www.nature.com/articles/s41467-022-28865-w
LIS	Local Interaction Score based on transform of PAEs from Kim, Hu, Comjean, Rodiger, Mohr, Perrimon. https://www.biorxiv.org/content/10.1101/2024.02.19.580970v1

For further details about the metrics and the original ipSAE implementation, refer to:

- Dunbrack Lab ipSAE repository: <https://github.com/DunbrackLab/IPSAE/tree/main>
- ipSAE publication: <https://doi.org/10.1101/2025.02.10.637595>

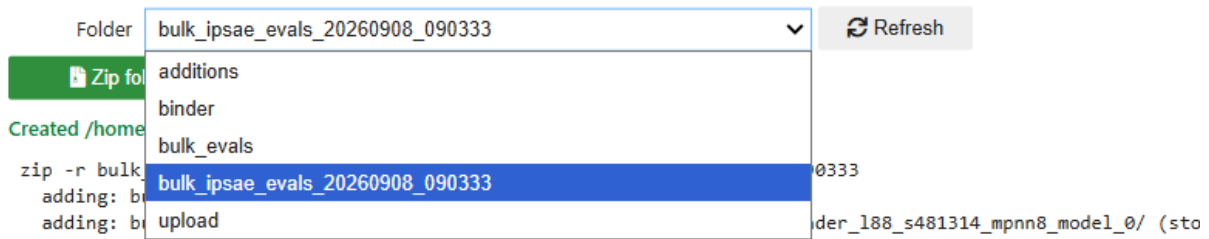


Model	Chr1	Chr2	Type	ipSAE	ipTM_af	pDockQ	LIS	Score_File
1 481314_mprn8_model_0	A	B	asym	0.812887	0.93	0.5119	0.7004	prn8_model_0_10_10.txt
2 481314_mprn5_model_0	A	B	asym	0.812041	0.93	0.5065	0.6958	prn5_model_0_10_10.txt
3 .ipynb_checkpoints				0.812				_10_bytes-checkpoint.txt
4 734871_mprn3_model_0	A	B	asym	0.808158	0.9	0.431	0.6398	prn3_model_0_10_10.txt
5 734871_mprn6_model_0	A	B	asym	0.791097	0.89	0.3884	0.6269	prn6_model_0_10_10.txt
6 s62602_mprn6_model_0	A	B	asym	0.780926	0.88	0.4781	0.5766	prn6_model_0_10_10.txt
7 450785_mprn2_model_0	A	B	asym	0.723004	0.85	0.4106	0.5229	prn2_model_0_10_10.txt
8 s62602_mprn1_model_0	A	B	asym	0.671827	0.83	0.3953	0.5505	prn1_model_0_10_10.txt
9 747661_mprn1_model_0	B	A	asym	0.546986	0.79	0.3618	0.4629	prn1_model_0_10_10.txt
10 450785_mprn5_model_0	B	A	asym	0.011191	0.24	0.2093	0.0677	prn5_model_0_10_10.txt
11 405521_mprn5_model_0	B	A	asym	0.010305	0.42	0.2206	0.0629	prn5_model_0_10_10.txt

4. IPSAE Output Export

To download a specific ipSAE output folder, open the **Zip Folder** panel and select the required output folder from the dropdown menu.

If the output was generated recently and not yet appear in the list, click **Refresh** to update the available folders.



The generated ZIP file will appear in the root directory of the project, visible in the **Explorer** panel on the left. Right-click and select **Download** to save it to your local computer.

